

Hand-in set 2 in FYS2130, spring 2026

General instructions. For all problems, you must clearly explain your reasoning and explicitly state any assumptions you make. Answers without justification will receive little or no credit. Where sketches or qualitative arguments are requested, they must be included and clearly labeled. Use the methods presented in the lectures.

Problem 1 (Coupled swings: resonance and energy transfer). We study the dynamics of two *identical, linearly coupled playground swings*. For small angular displacements, each swing can be modeled as a damped harmonic oscillator. A weak coupling between the swings allows energy to be transferred from one to the other. Only the first swing is driven by an external periodic force.

The equations of motion are

$$\begin{aligned}\ddot{\theta}_1 + 2\gamma\dot{\theta}_1 + \omega_0^2\theta_1 + \kappa(\theta_1 - \theta_2) &= \frac{F_0}{mL} \cos(\omega t), \\ \ddot{\theta}_2 + 2\gamma\dot{\theta}_2 + \omega_0^2\theta_2 + \kappa(\theta_2 - \theta_1) &= 0,\end{aligned}$$

where γ is the damping coefficient, $\omega_0 = \sqrt{g/L}$ is the natural frequency of each swing, κ measures the coupling strength, F_0 is the driving-force amplitude, and ω is the driving frequency.

- a) Give two examples of coupled oscillators from physics, engineering, or everyday life. What does it mean, physically, for two oscillators to be *coupled*?
- b) Only the first swing is driven. Explain in words how the motion of the second swing depends on:
 - the coupling strength κ ,
 - the driving frequency ω ,
 - the damping coefficient γ .
- c) Rewrite the equations as a system of four first-order differential equations for $\theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2$. Solve the system numerically using

$$L = 2.5 \text{ m}, \quad g = 9.81 \text{ m/s}^2, \quad m = 30 \text{ kg}, \quad F_0 = 15 \text{ N}, \quad \kappa = 0.8, \quad \gamma = 0.15 \text{ s}^{-1}.$$

Use the initial conditions

$$\theta_1(0) = 0.3, \quad \dot{\theta}_1(0) = 0, \quad \theta_2(0) = 0, \quad \dot{\theta}_2(0) = 0.$$

Plot $\theta_1(t)$ and $\theta_2(t)$. Compute and plot the energy of each swing as a function of time and describe how energy is exchanged between them.

- d) Vary the driving frequency ω and determine the steady-state amplitudes of both swings. At which frequencies does resonance occur? How do the coupling κ and damping γ modify the resonance behavior compared to a single driven oscillator?

- e) In the absence of external driving, the coupled system has two intrinsic frequencies,

$$\Omega_- = \omega_0, \quad \Omega_+ = \sqrt{\omega_0^2 + 2\kappa}.$$

If only one swing is initially displaced, the motion can be written as a superposition of the two normal modes, leading to *beats* with beat frequency

$$\Omega_{\text{beat}} = \Omega_+ - \Omega_-.$$

Demonstrate this behavior numerically and explain how the coupling strength κ and damping coefficient γ affect the beat pattern over time.

Problem 2 (Fourier analysis). Consider the 2π -periodic function defined on $(-\pi, \pi)$ by

$$f(x) = \begin{cases} 0, & -\pi < x < -\frac{\pi}{2}, \\ 1, & -\frac{\pi}{2} < x < \pi. \end{cases}$$

- a) Plot the function over three periods and indicate clearly the points where the function has jumps.
- b) The Fourier series of the 2π -periodic function $f(x)$ defined on $[-\pi, \pi]$ is defined as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx).$$

Using

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx,$$

evaluate the integrals analytically and derive explicit expressions for a_0 , a_n , and b_n .

- c) Represent the signal in the frequency-domain by plotting the magnitude of the Fourier coefficients $|c_n| = \sqrt{a_n^2 + b_n^2}$ as function of wavenumber n .
- d) Plot the partial sums of the Fourier series and describe how the approximation behaves near the discontinuities.